DEPI SOFTWARE TESTING TRACK — FINAL PROJECT

Project Plan

Work Breakdown Structure (WBS), 8-Week Execution Schedule, Project Milestones, and Technology Resource Plan

Project	Banking Transaction System	**Document**	Project Plan
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Part 1. Work Breakdown Structure (WBS)

The project scope is deconstructed into six structured execution phases aligned with the core banking system modules.

Phase	System Modules	Key Technical & Quality Deliverables
Phase 1: Project Setup & Architecture	Foundation & Database Infrastructure	• Initialize GitHub repository, branch protection rules, Spring Boot 3.5.x skeleton, Maven configuration, and development seed data. • Validate SQLite connection pooling (single-connection pool), Write-Ahead Logging (WAL) mode, whole-piaster BIGINT storage, and immediate transaction locking via an early architecture spike.
Phase 2: Authentication & User Management	Authentication (AUTH) & Customer Profile (CUS)	• Implement visitor registration, login/logout endpoints, stateful HTTP session handling, and stateless HTTP Basic API authentication. • Configure BCrypt password hashing, 5-attempt / 15-minute login lockout, 15-minute idle session timeout, Spring Security filter chains, and server-side Role-Based Access Control (CUSTOMER, ADMIN). • Develop customer profile management views and the account overview dashboard.
Phase 3: Core Banking Operations & Ledger	Accounts (ACC) & Transactions (TXN)	• Build object-oriented domain model classes for Savings, Checking, and Fixed Deposit account types with Luhn 12-digit account number generation. • Implement atomic deposit, withdrawal, and transfer service methods with rollback guarantees and idempotency key support. • Enforce single/daily/monthly withdrawal limits, Savings minimum balance (1,000 EGP), Checking overdraft (-1,000 EGP), Fixed Deposit lock-in rules, dynamic inter-customer transfer fees (0.1%, min 0.50 EGP, max 20.00 EGP), and append-only ledger entries.

Phase	System Modules	Key Technical & Quality Deliverables
Phase 4: Reporting, Administration & Scheduling	Statements (STM), Admin (ADM), Jobs (JOB) & Audit (AUD)	• Create paginated account statement views with newest-first ordering, post-transaction running balance, date/type filtering, and UTF-8 CSV export. • Build administrator dashboard metrics (total reserves, customer count, today's transactions/fees), customer search, balance-range account search, richest customer query, account freeze/unfreeze endpoints, and security/financial audit logging. • Implement scheduled background jobs (and manual admin triggers) for monthly savings interest accrual and fixed deposit maturity processing.
Phase 5: Automated & Security Testing	Unit, API, UI & Security Verification	• Develop JUnit 5 and Mockito unit tests for all domain business rules and MockMvc integration tests for REST controllers. • Build automated Postman API test collections (including P95 < 500 ms latency checks) and Selenium 4 WebDriver UI test scripts using Page Object Model and permanent HTML element IDs. • Execute dedicated Application Security test suites led by the Tech Lead & AppSec Engineer targeting OWASP Top 10 risks (MockMvc security tests, IDOR 404 enforcement, privilege escalation, lockout, SQLi, XSS, CSV injection, and CSRF).
Phase 6: Quality Assurance & Final Deliverables	Static Analysis, Coverage & Release Readiness	• Conduct static security analysis (SonarLint / SpotBugs) and OWASP Dependency-Check vulnerability scanning led by the Tech Lead & AppSec Engineer. • Triage and resolve defects logged in Jira, verify zero open Critical/High defects, generate JaCoCo code coverage and OpenAPI/Swagger documentation reports, and conduct full demo rehearsals.

Part 2. Timeline & Milestones

The 8-week execution schedule defines weekly engineering goals, dependencies, and formal project milestones.

2.1 Eight-Week Execution Schedule

Week	Primary Focus Area	Planned Weekly Activities & Outputs
Week 1	Project Foundations & Planning	Finalize Project Proposal, Project Plan, Task Assignment & Roles, KPIs, and Risk Assessment. Set up GitHub repository, database schema, SQLite WAL/piaster spike validation, and Master Test Plan initiation.
Week 2	Requirements & Core Skeleton	Complete Requirement Gathering specification (Stakeholders, User Stories, FRs, NFRs, Business Rules), Master Test Plan & RTM, domain entity classes, initial JUnit 5 unit tests, user registration, authentication, and RBAC.
Week 3	Core Banking Engine & APIs	Implement account opening/closing, deposits, withdrawals, internal/inter-customer transfers with fee quotes, atomic ledger logging, REST API endpoints, Thymeleaf transaction forms, and standardized error handling.

Week	Primary Focus Area	Planned Weekly Activities & Outputs
Week 4	Statements, Admin & UI Freeze	Deliver paginated account statements, date/type filters, CSV export, admin dashboard metrics, customer/account search, account freeze/unfreeze, audit log viewer, scheduled interest jobs, and freeze all HTML DOM element IDs.
Week 5	Test Automation & Security Suite	Develop functional Selenium 4 UI and Postman API suites (by QA Lead), alongside MockMvc & OWASP Top 10 security test suites covering IDOR 404, lockout, RBAC, CSRF, XSS, and SQLi (by Tech Lead & AppSec Engineer).
Week 6	Static Testing & Bug Fixing	Perform static security analysis (SonarLint / SpotBugs) and OWASP Dependency-Check scanning (by Tech Lead & AppSec), defect resolution in Jira, and full end-to-end functional regression testing.
Week 7	Documentation & Dry Runs	Verify OpenAPI/Swagger documentation, compile Jira test-activity and defect reports, generate JaCoCo coverage reports, verify all KPI targets, and conduct two full live demo rehearsals.
Week 8	Buffer & Final Defense	Schedule buffer for final polish, backup demo video recording, environment verification on primary and secondary laptops, and final project presentation and defense.

2.2 Key Project Milestones

Milestone	Target Timing	Milestone Title & Sign-Off Criteria
Milestone 1	End of Week 1	Project Foundations & Governance Sign-Off: Project Proposal, Project Plan, Task Assignment Matrix, KPIs, Risk Assessment Plan, and SQLite architecture spike verified.
Milestone 2	Weeks 2 – 3	Requirements Sign-Off & Core Banking MVP: Requirement Gathering specification, Master Test Plan & RTM, Authentication, and complete Transaction & Ledger engine with fee calculation verified.
Milestone 3	End of Week 4	Feature Complete & UI Element ID Freeze: Statements, CSV export, Admin dashboard, Audit logging, Scheduled jobs completed, and all HTML DOM IDs frozen for Selenium automation.
Milestone 4	End of Week 7	Code Quality & Test Sign-Off: 100% requirements test coverage, all planned automated and manual test suites executed, static analysis clean, and zero open Critical or High-severity defects.
Milestone 5	End of Week 8	Final System Delivery & Presentation: Rehearsed local live demonstration, complete test evidence package, and final team defense delivered.

Part 3. Resource & Technology Plan

Standardized technology stack, development frameworks, quality assurance tools, and runtime environment.

Resource Category	Tools, Frameworks & Environment Specifications
Backend Framework	Java 17 (or higher), Spring Boot 3.5.x (Spring Web, Spring Data JPA, Spring Security, Jakarta Bean Validation), Apache Maven (Maven Wrapper ./mvnw)

Resource Category	Tools, Frameworks & Environment Specifications
Database & Storage	SQLite 3 (embedded single-file relational database), xerial sqlite-jdbc driver, Hibernate Community Dialect for SQLite, DB Browser for SQLite (Configured in WAL mode with BIGINT piaster storage)
Frontend & User Interface	Server-side rendered Thymeleaf view templates, HTML5 with permanent automation IDs, responsive CSS3 / Bootstrap 5 styling
Testing & Coverage Tools	JUnit 5 (Jupiter), Mockito, Spring Boot Test / MockMvc, Selenium 4 WebDriver (Page Object Model with Chrome), Postman & Newman Runner, JaCoCo Code Coverage
Quality & Process Governance	Git & GitHub (protected branches, mandatory Pull Request reviews), Jira (Agile sprint board, RTM linkage, defect tracking), SonarLint / SpotBugs, OWASP Dependency-Check, Swagger / OpenAPI 3
Hardware & Runtime Host	Standard developer workstations (Windows 11 / Linux / macOS) running JDK 17+ and local embedded database environment requiring zero external server dependencies